

MACHINE LEARNING INTERNSHIP

TASK 1: IRIS CLASSIFICATION

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Colab Notebook link: [Task 1 Internspark ML.ipynb](#)

1. PROJECT OVERVIEW

The objective of this project is to build a machine learning classification model that predicts the species of an iris flower using four features: sepal length, sepal width, petal length, and petal width. The project demonstrates the complete machine learning workflow including exploratory data analysis, visualization, model training, evaluation, and model saving.

2. DATASET INFORMATION

Dataset Name: Iris Classification Dataset

Features:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

Target Variable:

- Species

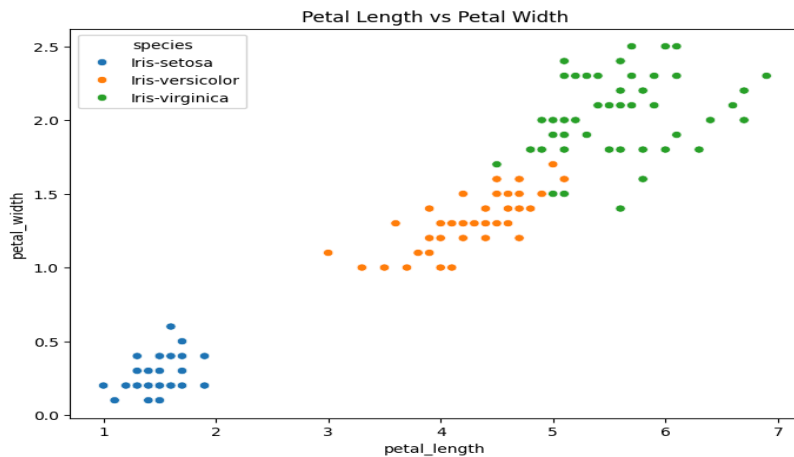
Classes:

- Iris-setosa
- Iris-versicolor
- Iris-virginica

Total Records: 150

3. DATA VISUALIZATION

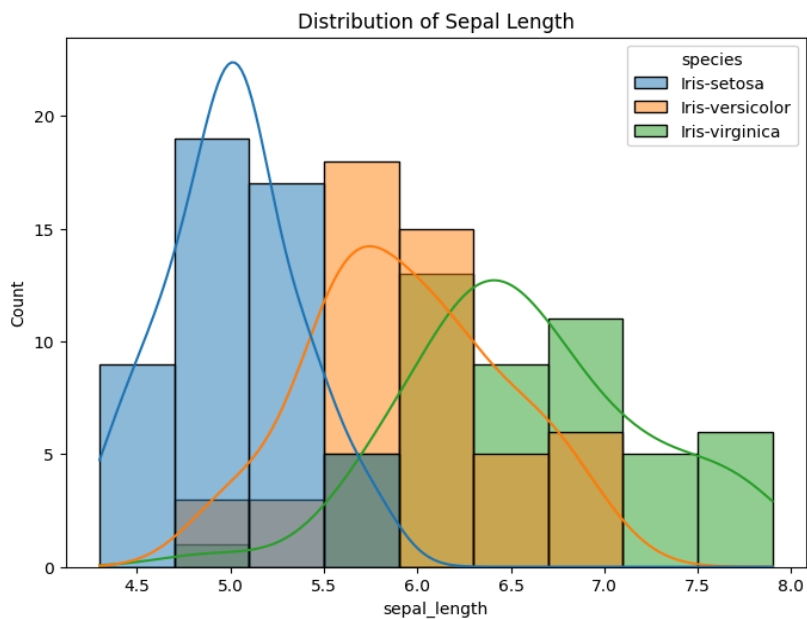
Figure 1: Scatter Plot (Petal Length vs Petal Width)



Observation:

The scatter plot clearly shows the separation between the three iris species. Iris-setosa is completely separated from the other species, while Iris-versicolor and Iris-virginica show slight overlap. Petal length and petal width are highly informative features for classification.

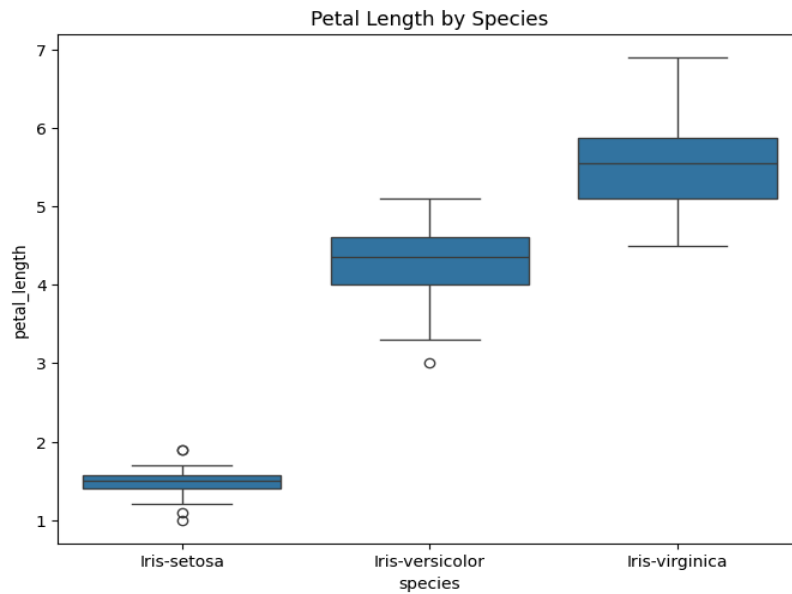
Figure 2: Histogram (Distribution of Sepal Length)



Observation:

The histogram displays the distribution of sepal length for each species. Iris-setosa generally has smaller sepal lengths, while Iris-virginica tends to have larger values. The distribution patterns help distinguish the species.

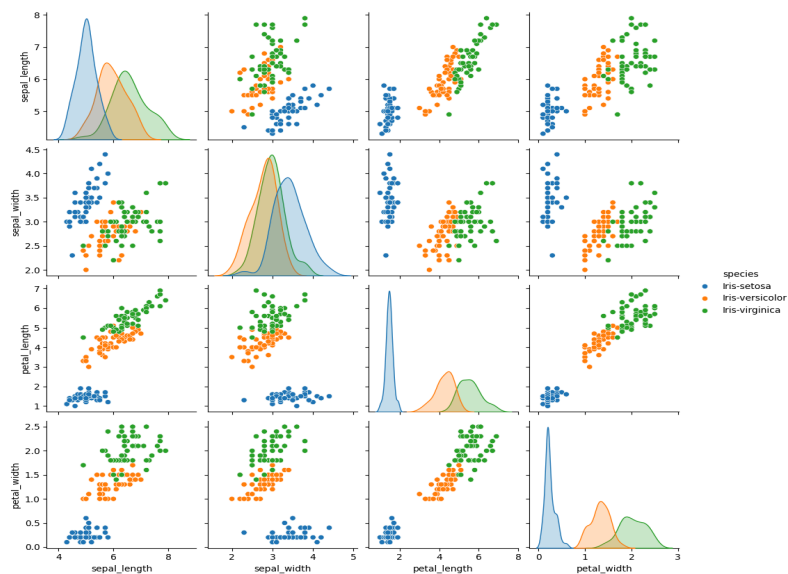
Figure 3: Boxplot (Petal Length by Species)



Observation:

The boxplot highlights the differences in petal length among the species. Iris-setosa has the smallest petal lengths, Iris-versicolor occupies the middle range, and Iris-virginica has the largest petal lengths. This feature strongly contributes to classification accuracy.

Figure 4: Pairplot



Observation:

The pairplot provides a comprehensive view of relationships between all features. The plots show strong class separability, especially for petal-related features. Distinct clusters can be observed for each species.

4. MODEL TRAINING

The following machine learning algorithms were trained and compared:

1. K-Nearest Neighbors (KNN)
2. Logistic Regression
3. Decision Tree Classifier

The dataset was split into training and testing sets using an 80:20 ratio.

5. MODEL PERFORMANCE

Accuracy Results:

K-Nearest Neighbors (KNN): 1.00

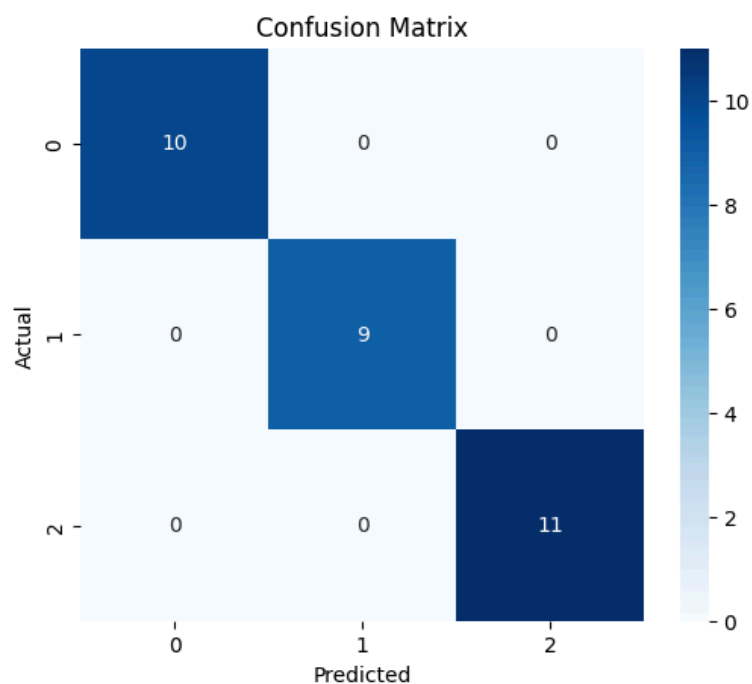
Logistic Regression: 1.00

Decision Tree Classifier: 1.00

Observation:

All three algorithms achieved 100% accuracy on the testing dataset. This is expected because the Iris dataset is highly structured and the classes are easily separable.

6. CONFUSION MATRIX



Observation:

The confusion matrix indicates that all test samples were classified correctly. No misclassifications occurred during testing.

7. MODEL SAVING

The best-performing model was saved using Joblib.

Saved File:
iris_model.pkl

This saved model can be loaded later for making predictions on new iris flower measurements.

8. GOOGLE DRIVE MODEL LINK

Model File Link:

<https://drive.google.com/file/d/1baxYoLYP2hmk6kXUCLoUiGVjMgU9D-nF/view?usp=sharing>

9.SUMMARY

This project focused on building a machine learning model to classify iris flower species based on flower measurements. The Iris dataset containing 150 samples and four numerical features was used for training and evaluation. Exploratory data analysis and visualization techniques were applied to understand the dataset and identify important patterns.

Several visualizations including scatter plots, histograms, boxplots, and pairplots were created to examine feature distributions and class separability. These visualizations demonstrated that petal length and petal width are particularly effective features for distinguishing between species.

Three classification algorithms were implemented and compared: K-Nearest Neighbors, Logistic Regression, and Decision Tree Classifier. All models achieved perfect classification accuracy on the testing dataset. The confusion matrix further confirmed that every test sample was classified correctly.

The best-performing model was saved as a deployment-ready artifact using Joblib. This project successfully demonstrated the complete machine learning workflow from data analysis to model deployment preparation and provided valuable practical experience in supervised machine learning.